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High-temperature wettability and interfacial interaction between CaO-MgO-Al₂O₃-SiO₂-Fe₂O₃ slag and MgO

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Abstract

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Highlights

- The high temperature interfacial wetting behavior of CMASF slag on different MgO crystal planes is revealed via AIMD simulation and DFT calculation.
- The CMASF/MgO(111) system shows the strongest interface interaction.
- In the CMASF/MgO(111) system, Ca diffuses the fastest, forming Ca₂SiO₄-like configuration.
- The surface energy can predict the interfacial wetting and reaction behavior qualitatively.

Abstract

The high-temperature wetting behavior of CaO-MgO-Al₂O₃-SiO₂-Fe₂O₃ (CMASF) slag on MgO refractories has been a focal point in metallurgical research. However, the interfacial interactions between CMASF and MgO at the atomic and molecular scales remain insufficiently understood. To address this problem, the high temperature interfacial wetting behavior of CMASF slag on different MgO crystal planes (i.e., (001), (110), (111) and (420)) was studied by *ab initio* molecular dynamics (AIMD) simulation. The results reveal a distinct sequence of wettability for CMASF on the different MgO crystal planes: CMASF/MgO(001) < CMASF/MgO(420) < CMASF/MgO(110) < CMASF/MgO(111). The CMASF/MgO(111) interface exhibits the strongest interfacial interaction. This is attributed to the synergistic effects of the highly electronegative oxygen-terminated MgO(111) surface and the pronounced mobility of Ca²⁺ within the slag. Collectively, these findings provide a theoretical foundation for the optimization of MgO refractories. Key design strategies include minimizing the exposure of the MgO(111) facet and enhancing resistance to Ca-induced erosion, thereby extending the material's high-temperature service life.

Graphical abstract

4. Conclusion

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgments

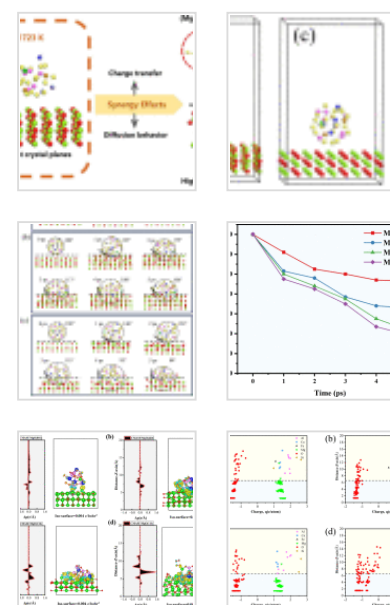
Appendix A. Supplementary data

Data availability

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Extras (1)



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Keywords

MgO refractories; CMASF slag; High-temperature wetting behavior

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1. Introduction

Magnesia refractories are extensively employed in high-temperature industrial applications [1], [2], [3], such as ironmaking and steelmaking, cement production, and non-ferrous metal processing. This is largely due to their high refractoriness, superior resistance to alkaline slags, and excellent thermal shock resistance [4], [5], [6], [7], [8]. MgO serves not only as a critical refractory lining in blast furnaces but also as a key additive in refining slags. The CaO-MgO-Al₂O₃-SiO₂-Fe₂O₃ (CMASF) slag system, in particular, has garnered significant attention. The high-alumina, low-calcium slag is characterized by its low cost, abundant availability, and superior properties [9], [10]. It plays a pivotal role during the smelting and continuous casting of high-aluminum

advanced high-strength steels by inhibiting the oxidation of dissolved aluminum and preventing excessive slag viscosity, thereby ensuring steel performance [11], [12], [13]. During high-temperature smelting, the refractory lining is subjected to aggressive slag attack. The combined effects of chemical corrosion and slag infiltration constitute the primary mechanisms leading to structural degradation and premature failure [14], [15], [16], [17]. To investigate the underlying mechanisms governing this interaction, numerous experimental studies have employed methodologies such as static corrosion tests and dynamic wetting experiments [18], [19]. However, experimentally capturing the real-time dynamic interactions between slag and MgO at high temperatures remains a significant challenge due to the complexity of the industrial smelting environment. This limitation hinders a comprehensive understanding of the reaction mechanisms, thereby impeding the rational optimization of refractories [20], [21]. Consequently, further exploration of the interfacial wetting behavior and atomic-scale interaction mechanisms between slag and MgO refractories holds significant theoretical and practical importance for improving refractory performance and prolonging its service life.

In recent years, advances in computational materials science have emerged as a powerful approach to elucidate microscopic reaction mechanisms [22], [23], [24]. *Ab initio* molecular dynamics (AIMD) simulations enable the analysis of thermodynamic and kinetic processes governing interactions directly at the atomic scale [25], [26], [27], [28], thereby offsetting the limitations inherent to experimental methodologies. For instance, Jiang et al. [29] studied the effect of MgO/Al₂O₃ ratio on the structural characteristics and properties of SiO₂-Al₂O₃-CaO-MgO slag system by AIMD. It was found that increasing the MgO/Al₂O₃ ratio can reduce the stability of [SiO₄]⁴⁻ and [AlO₄]⁵⁻ tetrahedron units, and decrease the relative proportion of bridge oxygen and non-bridge oxygen atoms, thus reducing the viscosity of slag, which is consistent with the experimental results. In addition, AIMD simulations have been used to study interfacial reaction. Fang et al. [30] studied the heterogeneous nucleation of liquid Al on the surface of MgO. The results show that the newly formed Al-layers terminating the substrates display structural splitting and possess atomic vacancies, being atomically rough, thereby weakening the pre-

nucleation phenomenon at the liquid/oxide interfaces. Niu et al. [31] explored the interfacial reaction behavior of alloys on ceramic surfaces and suggested using the density functional theory (DFT) calculated interfacial binding energy to explain the trend of wettability in experimental results. These computational examples thereby provide a reliable methodology for obtaining atomic-level insights into interactions at interfaces efficiently.

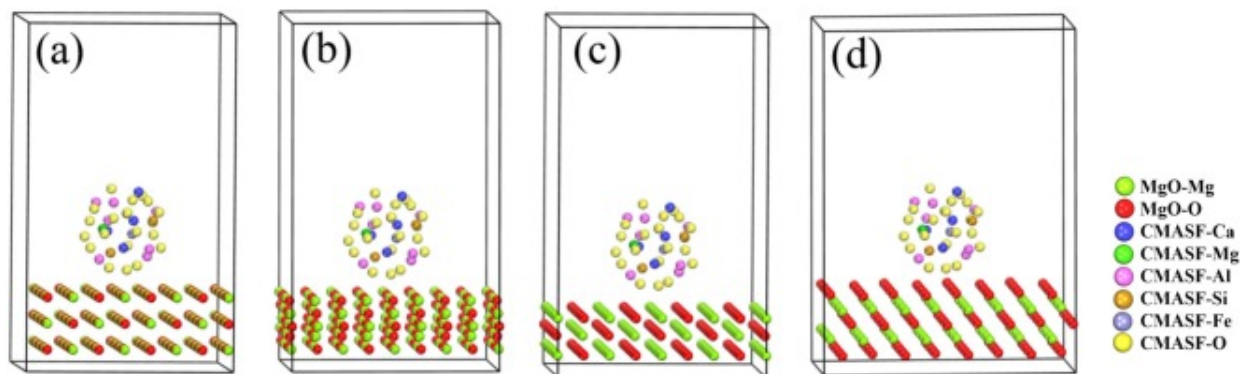
While extensive research has focused on the structural properties of slags, investigations regarding the high-temperature interfacial behavior between complex multi-component slags and refractories remain limited. To bridge the research gap regarding the atomic-scale dynamic wetting and interfacial reaction mechanisms of such complex high-alumina slag systems, this study selects the CMASF system to investigate its interaction mechanism with magnesia refractories. Given that practical MgO refractories exhibit complex microstructures with diverse exposed crystallographic orientations, this study employs AIMD to systematically simulate the high-temperature wetting behavior of CMASF slag on different MgO crystal planes. Investigating distinct crystal planes allows us to indirectly reflect the impact of MgO crystallinity and density on interfacial reactions and refractory performance under practical service environments. To systematically elucidate the atomic-scale corrosion mechanism of CMASF slag on different MgO crystal planes and provide a theoretical basis for designing more stable, corrosion-resistant refractories, DFT calculations were performed to analyze the surface energy of MgO, interfacial reaction, interaction energy and electron transfer characteristics between CMASF slag and different MgO crystal planes.

2. Model construction and calculation methods

The initial cluster model of CMASF slag (Ca₄MgAl₇Si₂FeO₂₁) was constructed according to the chemical composition of experiment [32], and its proportion was close to the chemical composition of the actual slag (Mass ratio of slag: CaO:MgO:Al₂O₃:SiO₂:Fe₂O₃=23.51:5.05:43.97:9.91:8.87, Fig. S1). At present, the atomic-scale dynamic wetting and reaction mechanisms between this complex high-alumina slag

and different MgO crystal surfaces remain unclear. Referring to the measurement of high temperature wetting angle in the experiment, the nano-droplet of approximate sphere for CMASF slag was selected [33], [34], [35]. After equilibrium by AIMD simulation, spherical morphology with an approximate radius of 8 Å is achieved for the CMASF slag, driven by the minimization of its total energy. The pre-equilibrated CMASF slag demonstrates marked energy convergence and structural stability (Fig. S2). Corresponding Bader charge values are approximate to those observed experimentally for the slag components [30], [36], thereby validating the physical rationality of the simulated CMASF slag model (Table S1).

In addition, after optimizing the MgO unit cell, super-cell models of MgO(001), MgO(110), MgO(420), and MgO(111) surfaces (Fig. S3 (a~d)) were constructed according to the dominant diffraction peaks of XRD [37], [38] (Fig. S4). The O terminated MgO(111) crystal plane was adopted owing to its lower surface energy [39], [40]. To further study the interaction between the CMASF slag and MgO with different crystal planes, the interface models were constructed and designated as: CMASF/MgO(001), CMASF/MgO(110), CMASF/MgO(420), and CMASF/MgO(111), as shown in Fig. 1(a~d). According to the relevant research, the interface spacing between slag and MgO is set to 2 Å [41], [42]. A vacuum layer of 20 Å is set above the slag to eliminate the periodic mirroring effect along the z-axis.



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Fig. 1. Interface models: (a) CMASF/MgO(001), (b) CMASF/MgO(110), (c) CMASF/MgO(420) and (d) CMASF/MgO(111). (In the slag, yellow is O atom, brown is Si atom, green is Mg atom, pink is Al atom, blue is Ca atom, and purple is Fe atom; in the MgO, red is O atom and green is Mg atom.)

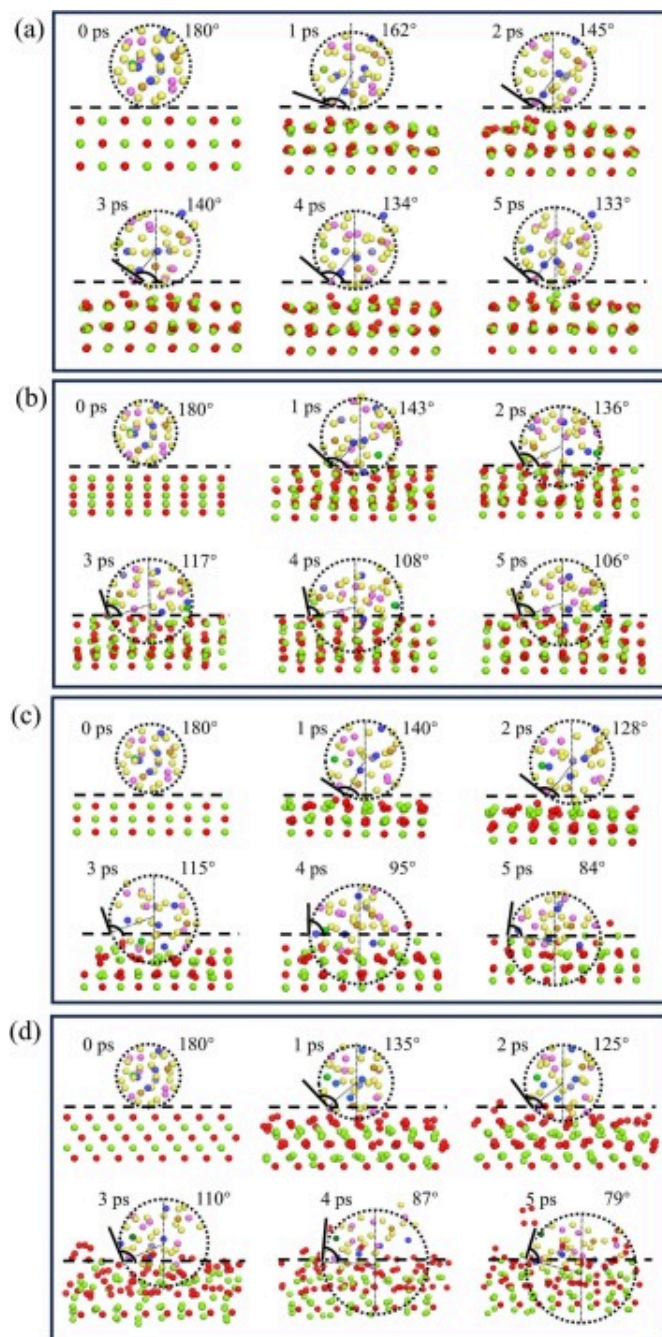
All DFT calculations and AIMD simulations are performed by employing the Vienna Ab Initio Simulation Package (VASP) [43], [44]. The Generalized Gradient Approximation (GGA) [45] with the Perdew-Burke-Ernzerhof (PBE) functional is used, and the Plane Wave-based Projected Augmented Wave (PAW) [46], [47] method is applied. AIMD simulations are conducted using the NVT ensemble and the Nosé-Hoover thermostat [48], [49]. Furthermore, all static DFT calculations are performed with spin-polarization turned on. Monkhorst-Pack special KPOINTS ($2 \times 2 \times 1$) are used to efficiently calculate interface models [50]. The time step is set as 2fs, and the total simulation time is 5ps. The simulated temperature was set to 1723K that approximately close to the high temperature environment in the actual production [51]. Given that the simulation temperature exceeds the Curie temperature of Fe ($\sim 1043\text{K}$), spin polarization was omitted from the AIMD simulations to balance computational efficiency and cost. The optimization parameters were set as follows: cutoff energy of the system was established at 450eV (This convergence criterion was fully satisfied, as evidenced by the cut-off energy test results in Fig. S5).

3. Result and discussion

3.1. Interfacial wetting behavior

Fig. 2(a~d) presents sequential snapshots (0~5ps) depicting structural evolution during AIMD simulations for the CMASF/MgO(001), CMASF/MgO(420), CMASF/MgO(110), and CMASF/MgO(111) systems, respectively. The tangent geometry method [52] was employed to determine the corresponding contact angles (See supporting information for details, Fig. S6). Fig. 2(a) reveals that CMASF slag maintains a similar spherical morphology on MgO(001), while the underlying MgO(001) crystalline structure exhibits

minimal degradation. No discernible interfacial and atomic diffusion occurs between CMASF slag and MgO(001) substrate. And the contact angle undergoes gradual reduction slowly while maintaining an approximately spherical droplet morphology. Contact angle decreases steadily from 180° (t=0 ps) to 133° after 5 ps of simulation at 1723K, indicating that CMASF slag is non-wetting on MgO(001).



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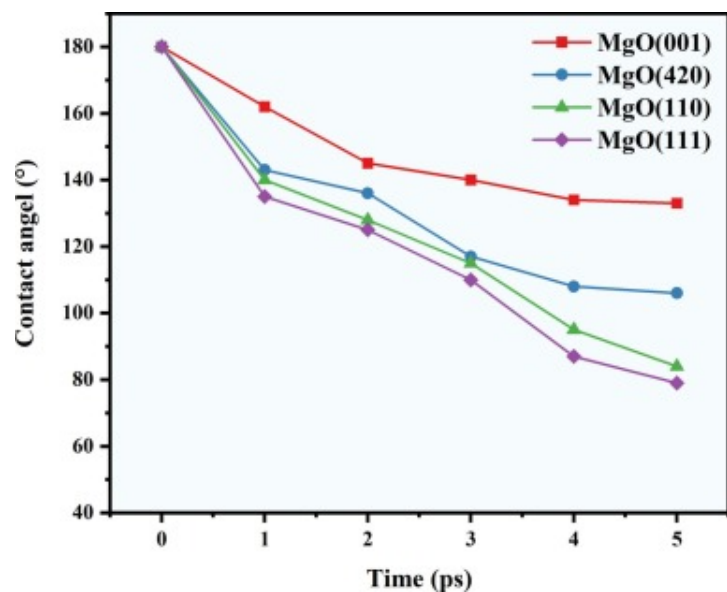
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Fig. 2. Contact angle snapshots of (a) CMASF/MgO(001), (b) CMASF/MgO(420), (c) CMASF/MgO(110) and (d) CMASF/MgO(111) during AIMD simulations.

As shown in Fig. 2(b~d), the structures of slag and MgO in CMASF/MgO(420), CMASF/MgO(110) and CMASF/MgO(111) systems change significantly with the increase of simulation time. Distinct wetting behaviors emerge for these systems, accompanied by progressive interfacial inter-diffusion between CMASF slag and MgO substrate at the interfaces. In the CMASF/MgO(420) system (Fig. 2(b)), the Ca atom from the slag begin diffusing into the MgO substrate at 1 ps. Subsequently, Al atoms migrate downward progressively, accompanied by contact angle reduction from 180° (0 ps) to 106° (5 ps). The CMASF/MgO(110) interface (Fig. 2(c)) exhibits more pronounced inter-diffusion, and Al, Ca, and O atoms from the CMASF slag diffuse into the MgO lattice, reducing the contact angle to 84° (5 ps) finally. Most severe interfacial inter-diffusion occurs in the CMASF/MgO(111) system (Fig. 2(d)), where the O, Si, Al, and Ca atoms from the CMASF slag migrate downward while O atoms of MgO diffuse upward, resulting in a final contact angle of 79° with continuous spreading of the CMASF slag across the MgO(111) substrate. Notably, the CMASF/MgO(001) and CMASF/MgO(420) systems exhibited incomplete inter-diffusion after 5 ps of AIMD simulations. Conversely, CMASF slag exhibits pronounced inter-diffusion and interfacial reaction with MgO(110) and MgO(111) surfaces.

In order to show the change of contact angle of CMAS slag in the four systems more intuitively, the change of contact angle with time was counted, as shown in Fig. 3. Results of contact angle reveal the following order for interfacial interaction strength: CMASF/MgO(111)>CMASF/MgO(110)>CMASF/MgO(420)>CMASF/MgO(001). Contact angle measurements across different interfacial systems reveal distinct wetting behaviors of CMASF slags on MgO with various crystallographic planes. Comparative analysis reveals the wetting behavior gradually comes to completion during 0–4 ps and approaches equilibrium during 4–5 ps. The contact angle of CMASF slag on MgO(111)

plummeted from 180° to 87° during 0–4 ps, demonstrating the fastest wetting transition and near-complete spreading on MgO(111). Conversely, CMASF slag showed the smallest wetting response on MgO(001), the contact angle only decreased from 180° to 134° during 0–4 ps, and quasi-spherical droplet morphology is maintained even at final equilibrium. During 4–5 ps, the contact angle of CMASF slag on MgO(001) and MgO(420) surfaces remained almost unchanged, whereas the MgO(110) and MgO(111) surfaces exhibit a clear tendency to continue the interfacial reactions. Correspondingly, the CMASF/MgO(001) system exhibits the largest equilibrium contact angle ($\theta=133^\circ$), while CMASF/MgO(111) system demonstrates the most pronounced wetting with contact angle of 79° at 5 ps. Povstenko et al. [53] demonstrated that crystallographic orientation significantly influences the wettability of the substrate surface contact angles, and the surfaces exhibiting lower atomic packing densities yield contact angles up to twice those observed on higher atomic density crystal plane, which is consistent with computational findings in this study.



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Fig. 3. Contact angle changes of CMASF slag on different MgO surfaces during AIMD simulation.

3.2. Interface electronic properties

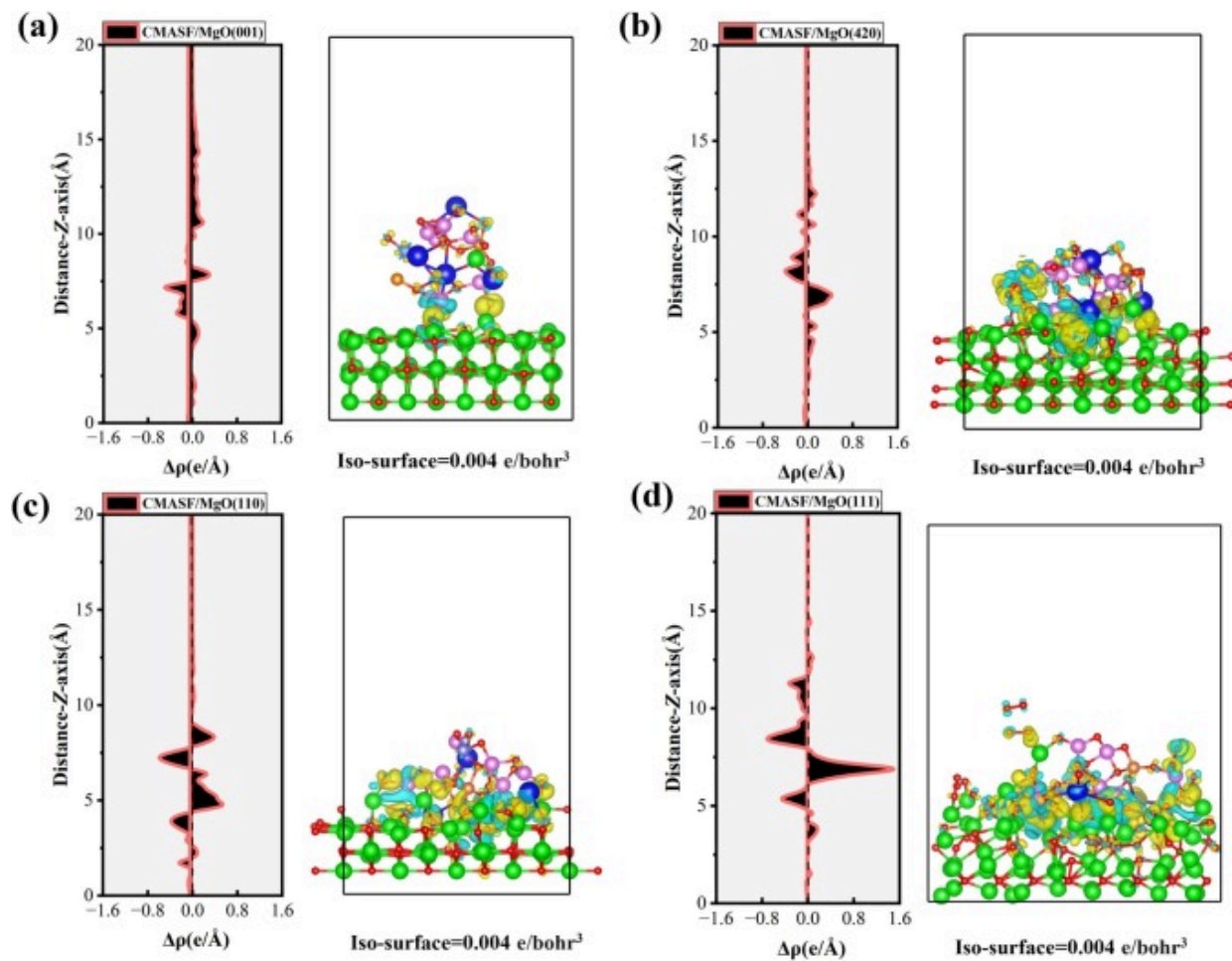
To further elucidate interfacial wetting behavior of CMASF on different MgO surfaces, we explore the electronic properties of the interfaces at 5ps by charge density difference, which intuitively reflects the interfacial interactions. The computation formula employed is outlined as follows:

$$\Delta\rho = \rho_{AB} - \rho_A - \rho_B \quad (1)$$

where ρ_{AB} represents the overall charge density, ρ_A represents the charge density of CMASF slag, ρ_B represents the charge density of MgO substrates, and $\Delta\rho$ can be expressed as the degree of charge transfer and accumulation at the interfaces.

Fig. 4(a~d) shows the 1D and 3D charge density difference of CMASF/MgO(001), CMASF/MgO(110), CMASF/MgO(420) and CMASF/MgO(111) interface models. The charge density difference reveals the degree of interaction and bonding strength between the constituents based on the electron cloud density. The yellow region represents electron accumulation, while the cyan region represents electron loss. More obvious electron accumulation and loss regions mean a greater degree of charge transfer, indicating stronger interface interaction. In the case of CMASF/MgO(001) and CMASF/MgO(420) interfaces, the charge density difference shows a relatively weak interaction. The limited electron accumulation and loss regions indicate that the interfacial interactions for these two systems are relatively mild. Compared with the CMASF/MgO(001) system, the yellow and cyan regions in the CMASF/MgO(420) system are larger, which indicates that the interface reaction of the CMASF/MgO(420) system is relatively stronger. At the CMASF/MgO(110) interface, the electron cloud density shows a more significant change than that in CMASF/MgO(420), indicating a stronger interfacial interaction. This change may be due to the obvious structural adjustment of the MgO(110) surface, resulting in additional bond breakage or reconstruction at interface, thereby enhancing the charge

transfer and interfacial interaction. The charge density difference of CMASF/MgO(111) interface shows the highest electron cloud redistribution with a wider range of yellow and cyan regions, indicating a large-scale charge transfer and stronger interface interaction. This is because the oxygen atoms exposed on the MgO(111) crystal plane are conducive to electron transfer and promote the interfacial inter-reaction. The 1D charge density difference further confirms these trends quantitatively. The integral area in CMASF/MgO(420) system is slightly larger than that in CMASF/MgO(001) system, and both of them are smaller than that in CMASF/MgO(110) system. However, the positive- and-negative lobes of CMASF/MgO(111) system give the largest area, indicating that the interfacial interaction is the strongest.



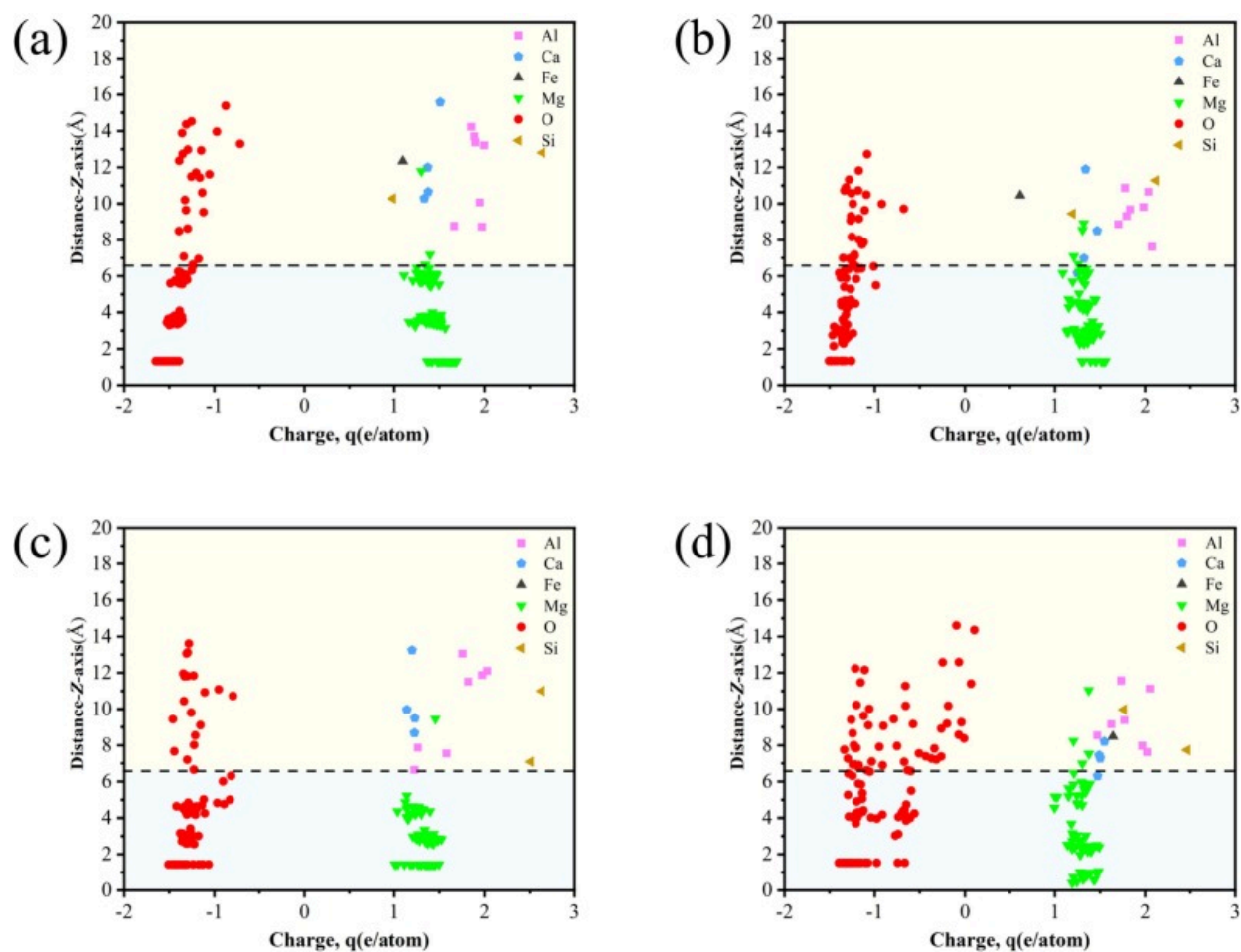
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Fig. 4. 1D and 3D charge density difference: (a) CMASF/MgO(001), (b) CMASF/MgO(420), (c) CMASF/MgO(110), (d) CMASF/MgO(111). The iso-surface value is 0.004 e/bohr³ (yellow and cyan represent charge accumulation and loss, respectively).

Bader charge analysis can be used for estimating the valence states of atoms by assessing the number of electrons in their surrounding environment [54], [55]. The Bader charges of the four systems at 5ps were calculated to further quantify the electron transfer across

the interfaces. Fig. 5 presents the Bader charge of each atom that distributed along the z-axis of each interface system at 5 ps, where dashed line demarcates the CMASF slag from the MgO substrates. The interfacial charge transfer behavior of CMASF slag exhibits remarkable disparities on diverse MgO crystal planes. The inter-diffusion between CMASF slag and MgO(001) is not obvious (Fig. 5(a)), and the charge distribution of each element in slag is more concentrated, indicating that the interface interaction is weak. The CMASF/MgO(420) system shows rapid down-diffusion of a small amount of Ca and Al atoms from the slag (Fig. 5(b)), which preferentially react with MgO and accumulate at the interface. The average Bader charge of Mg and O atoms at the CMASF/MgO(001) and CMASF/MgO(420) interfaces changes little (Table S2), which is consistent with the approximate valence state of the ion model (Mg^{2+} , O^{2-}) [56]. This indicates that the surface structure of MgO(001) and MgO(420) is relatively intact, and no serious slag erosion is received. A more pronounced inter-reaction is observed at CMASF/MgO(001) interface (Fig. 5(c)), where a greater quantity of Al, Si, and Ca atoms from CMASF slag migrate to and accumulate at the interface. The Bader charge distribution of the atoms in slag and MgO at the interface becomes more dispersive, indicating that the interface interaction is stronger.



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Fig. 5. The Bader charge profiles of (a) CMASF/MgO(001), (b) CMASF/MgO(420), (c) CMASF/MgO(110) and (d) CMASF/MgO(111).

The strongest interaction occurs on the CMASF/MgO(111) interface (Fig. 5(d)), and more Ca, Al and Si atoms accumulate at the interface, accompanied by the phenomenon of atomic mutual diffusion between CMASF slag and MgO(111). As illustrated in Fig. 2, Fig. 5, some Ca atoms from slag diffuses partially to sub-surface of the MgO(111), and Mg ions

dissolve from the MgO(111) to the slag, damaging the surface structure of MgO(111) substrate. Consequently, the CMASF/MgO(111) interface displays the broadest Bader charge distribution, indicating the charge transfer is more obvious and the interfacial interactions between CMASF slag and MgO(111) is the most intense. Thus, MgO(111) is the highest reactivity among all the surfaces. Interfacial Bader charge analysis yields values of +1.50e/Ca, +1.80e/Al, and +2.11e/Si (Table S2), corresponding to characteristic valence states of Ca²⁺, Al³⁺, and Si⁴⁺, respectively. Based on the Bader charge analysis, it is speculated some stable phases similar to aluminate and silicate are formed between CMASF slag and the MgO(111) substrate. Therefore, a stronger interfacial reaction leads to a smaller contact angle, thereby improving wettability. Conversely, a weaker interfacial reaction induces minimal alteration to the surface structure of MgO(001) surface.

3.3. Correlation of surface energy and contact angle

To quantitatively establish the relationship between wetting behaviors and interfacial interactions, the interfacial interaction energies (E_{Int}) and surface energies ($E_{surface}$) were calculated using Eqs. (2), (3), respectively.

$$E_{Int} = \frac{E_{System} - E_{Slag} + E_{Substrate}}{S} \quad (2)$$

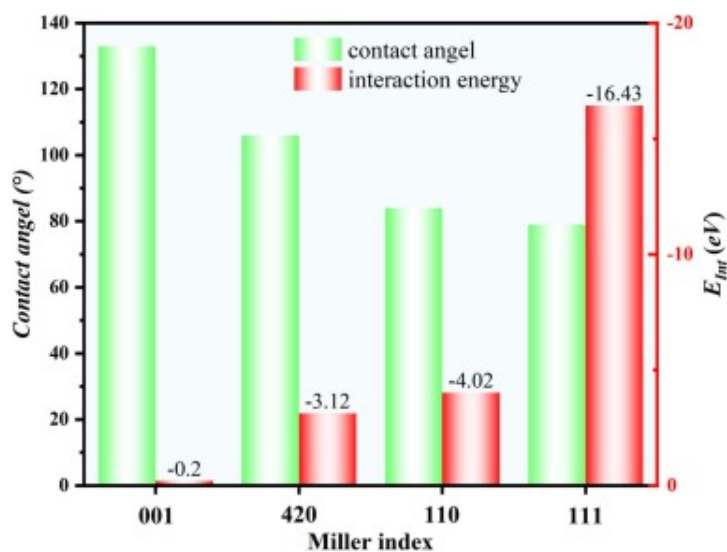
where E_{Int} is the interaction energy between CMASF slag and MgO (at 5ps); E_{Slag} is the energy of CMASF slag; $E_{Substrate}$ is the energy of MgO; E_{system} is the energy of interface model system.

$$E_{surface} = \frac{E_{total} - nE_b}{2S} \quad (3)$$

where E_{total} is the total energy of the MgO surface, n is the total number of MgO unit cell contained in the surface, E_b is the energy of MgO unit cell, and S is the surface area, $2S$ represents the slab model has upper and lower surface.

Fig. 6 depicts the relationship between contact angles and interfacial interaction energies for CMASF/MgO systems at 5ps. The CMASF/MgO(111) interface exhibits the highest interaction energy and lowest contact angle ($\theta=79^\circ$), while CMASF/MgO(001)

demonstrates the lowest interaction energy with the largest contact angle ($\theta=133^\circ$). This inverse correlation likely originates from enhanced surface chemical activity in substrates with strong interfacial binding (such as MgO(111)), contrasted with reduced chemical activity in low-energy surfaces (such as MgO(001)) [57], [58].



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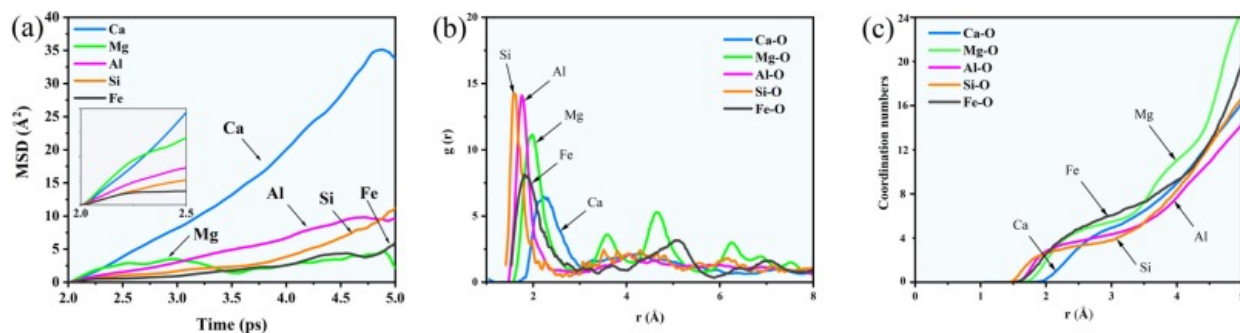
Fig. 6. Relationship between the contact angles and the interaction energies of CMASF/MgO at 5 ps.

While AIMD simulations effectively visualize the wetting dynamics of molten slag on distinct interfaces, the computational expense remains substantial. To evaluate whether initial surface energies can predict reaction behavior and wettability relationships, the surface energies of four MgO crystallographic planes were calculated. As shown in Fig. S7, the calculated surface energies reveal that MgO(001) exhibits the lowest surface energy (0.80J/m^2), whereas MgO(111) demonstrates the highest (4.53J/m^2). The surface energies follows: $E_{\text{surface-MgO}(001)} < E_{\text{surface-MgO}(420)} < E_{\text{surface-MgO}(110)} < E_{\text{surface-MgO}(111)}$. It can be seen that the trend of surface energies (E_{surface}) is completely consistent with the trend of interfacial

interaction energies (E_{Int}) for different MgO surfaces. That is to say, the surface energy variation trend of MgO surfaces correlates positively with interfacial interaction energies while exhibiting an inverse relationship to change trend of contact angle (Fig. S7). This systematic correlation indicates that MgO surfaces with high chemical activity exert enhanced interfacial affinity toward CMASF slag, inducing rapid wetting and spreading on the substrate. Specifically, higher surface energy corresponds to greater chemical reactivity and greater interfacial interaction energy, which corresponds to smaller contact angle, better wettability and more intense interfacial reaction. These observations align precisely with simulation trends, confirming that surface energy with less time-consuming calculations can provide predictive capability for interfacial wetting and reaction behavior. Consequently, this methodology establishes significant structure–activity relationships with broader implications for analogous interfacial systems and provide a valuable insight for predicting the interfacial wetting behavior by high-throughput calculation in the future.

3.4. Interfacial reaction mechanism of CMASF/MgO(111)

AIMD simulations reveal pronounced interfacial wetting and spreading of CMASF slag on the MgO(111) surface, accompanied by significant atomic inter-diffusion. The MgO(111) surface shows strong reactivity compared with the other three surfaces. Focusing on the interfacial reaction between CMASF and MgO(111) can help us to understand the reaction mechanism efficiently so as to better guide the experiments. To elucidate the underlying mechanisms governing rapid wetting and interfacial reaction for CMASF/MgO(111) system, mean squared displacement (MSD), radial distribution function (RDF), and coordination number (CN) analyses of the CMASF slag during 2–5 ps were calculated by the following formula, respectively (Eqs. (4), (5), (6)). The corresponding computed results are presented in Fig. 7.



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Fig. 7. (a) Mean-square displacement of CMASF slag on MgO(111), (b) radial distribution function, and (c) coordination number for Ca-O, Mg-O, Al-O, Si-O, and Fe-O during 2–5 ps.

The mean squared displacement (MSD) reflects the extent of diffusion of different elements and is defined by Eq. (4).

$$\langle \Delta r^2 t \rangle = \langle r t - r_0^2 \rangle \quad (4)$$

Where $r(t)$ is the position vector of an atom at time t ; $|r(t) - r(0)|^2$ is the squared displacement of that atom from its initial position; the angle brackets $\langle \rangle$ denote an ensemble average over all atoms; and $\langle \Delta r^2(t) \rangle$ represents the MSD of the system.

The radial distribution function (RDF) is calculated by Eq. (5).

$$g(r) = \frac{n(r, \Delta r)}{4\pi r^2 \Delta r \rho_0} \quad (5)$$

RDF usually denoted by $g(r)$, is a function that describes the ratio of the probability density of other atoms (or molecules) to the probability density under random distribution at a distance r from a reference atom (or molecule). $n(r, \Delta r)$ is the average number of atoms in the spherical shell from r to $r + \Delta r$ in the statistical time. $\rho_0 = N/V$, which is the average number density of the system (N is the total number of atoms, V is the volume of the system).

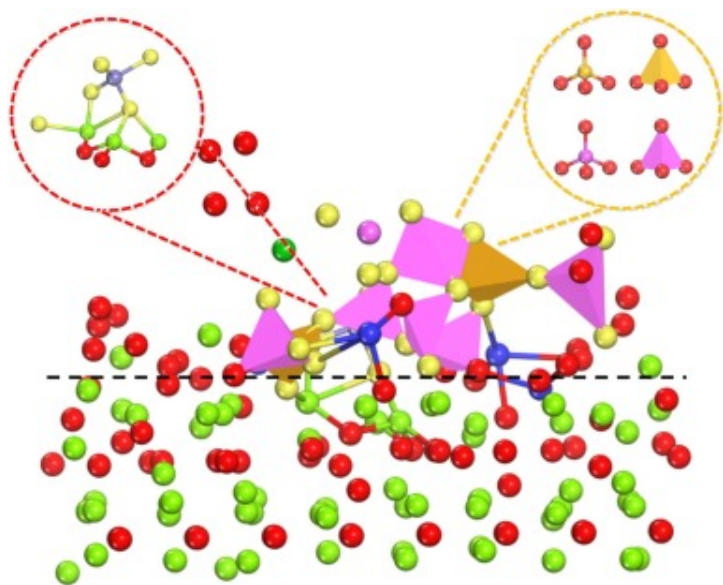
Accordingly, the coordination number (CN) is calculated by Eq. (6).

$$\text{CN} = 4\pi\rho_0 \int_0^{r_{\min}} g(r)r^2 dr \quad (6)$$

Mean squared displacement (MSD) calculations demonstrate increasing atomic mobility with prolonged CMASF/MgO(111) interaction, and the diffusivity of atoms in CMASF slag during 2.0–2.5 ps follows: Ca>Mg>Al>Si>Fe. Radial distribution function (RDF) analysis reveals pronounced short-range ordering structures, with Si-O and Al-O bonds exhibiting stronger peak intensities than Mg-O and Ca-O bonds. Consequently, these covalent networks (Si-O and Al-O) exhibit greater bond stability, impeding metal ion migration, especially for Mg ion in the last 2 ps. The interfacial stable phase that possibly form around 2.5 ps constrains Mg mobility within the CMASF slag. Between 3–5 ps, the mean squared displacement (MSD) of Mg atoms exhibits plateauing state with intermittent decreases, another possible reason is that the Mg atoms from MgO competes with the Mg in the slag for diffusion. Coordination number (CN) calculations indicate tetrahedral coordination for both Si (CN=4) and Al (CN=4), corresponding to [SiO₄]⁴⁻ and [AlO₄]⁵⁻ structural units (Fig. 7(b)). This tetrahedral confinement restricts Si/Al mobility, causing their migration only through complex anion group transport. The diffusion of Fe atoms with smallest MSD is the slowest, implying that similar iron-containing intermediate phase may form at the interfaces.

To identify phases formed during interfacial reactions, bond lengths for Ca-O, Si-O, Al-O, Si-Ca, and Al-Ca nearby [SiO₄]⁴⁻ and [AlO₄]⁵⁻ tetrahedral units were measured for geometric configuration at 5 ps (Fig. 8). These measurements were compared against reference bond lengths in crystalline Ca₂SiO₄ (C₂S) and CaAl₂O₄ (CA₂) structures (Fig. S8), as shown in Table S3. The measured bond lengths exhibit closer alignment with Ca₂SiO₄ crystal parameters. This indicates that the high mobility Ca atoms migrate to the interface and bind to the terminal O atoms of the adjacent [SiO₄]⁴⁻/[AlO₄]⁵⁻ groups to preferentially form a Ca₂SiO₄-like configuration (Fig. 8), which is consistent with the experimental results of Park et al. [59], who observed the formation of Ca₂SiO₄ ring-like structures around MgO inclusions. The formation of this structure indirectly facilitates the interfacial

reaction. Similarly, Fe atoms form (Mg, Fe)O intermediate phase with the MgO (Fig. 8), leading to the higher coordination number and the weakest diffusion ability of Fe atoms. Therefore, we need to pre-treat the slag and regulate the element content in the slag to achieve precise adjustment of the interface interaction strength and atomic diffusion rate requirements, thereby improving the high temperature stability of the material.



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Fig. 8. Snapshots of CMASF/MgO(111) at 5ps. The yellow circle on the right encloses the $[\text{SiO}_4]^{4-}$ and $[\text{AlO}_4]^{5-}$ tetrahedral structures, and the red circle on the left encircles the (Mg, Fe)O intermediate phase.

4. Conclusion

The high temperature wettability and interfacial interactions of CMASF/MgO(001), CMASF/MgO(110), CMASF/MgO(420) and CMASF/MgO(111) were studied by AIMD simulations and DFT calculations. Key findings demonstrate that:

- (1) Interfacial interaction strength between CMASF slag and MgO substrates follows in the order:
CMASF/MgO(111)>CMASF/MgO(110)>CMASF/MgO(420)>CMASF/MgO(001).
The higher the surface energy, the stronger the wettability and the greater the interfacial interaction energy of the corresponding interface.
- (2) For CMASF/MgO(001) and CMASF/MgO(420), the interface interaction is weak and inter-diffusion is not obvious, accompanied by a smaller amount of charge transfer at the interface. Relatively more obvious interface interaction occurs at the CMASF/MgO(110) interface. The CMASF/MgO(111) interface shows the strongest interface interaction, violent inter-reaction, and the greatest number of charge transfers.
- (3) During AIMD simulation of CMASF/MgO(111) interface, Ca diffuses fastest and preferentially bonds with terminal O atoms in adjacent [SiO₄]⁴⁻ and [AlO₄]⁵⁻ structural units, exhibiting possibility to form Ca₂SiO₄-like configuration. The Fe atom in slag may form (Mg, Fe)O intermediate phase with the MgO, increasing its coordination number, and making the diffusivity of Fe atom lowest.

Consequently, experimental efforts should prioritize suppressing the exposure of MgO(111) surfaces while fostering the formation of thermodynamically stable surfaces, such as MgO(001), to extend the service life of refractory materials. The results reveal the crystal facet dependence of interfacial wetting and interaction behaviors in the CMASF/MgO system, providing theoretical support for enhancing refractory performance by tailoring the substrate crystal orientation.

CRediT authorship contribution statement

Yixiang Qi: Writing – original draft, Formal analysis, Data curation. **Xingli Liu:** Writing – review & editing, Visualization, Software. **Long Dong:** Writing – review & editing,

Visualization, Investigation. **Liang Huang:** Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Haijun Zhang:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Shaowei Zhang:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Zhijun Dong:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following are the Supplementary data to this article:

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Supplementary Data 1.

Data availability

Data will be made available on request.

References

- [1] C. Liu, X. Gao, S. Ueda, M. Guo, S.-Y. Kitamura
Composition changes of inclusions by reaction with slag and refractory: a review
ISI Int., 60 (9) (2020), pp. 1835-1848
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] W.E. Lee, S. Zhang
Melt corrosion of oxide and oxide–carbon refractories
Int. Mater. Rev., 44 (3) (1999), pp. 77-104
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] B. Ma, X. Ren, Effects of key factors on the densification and grain growth behaviour of sintered magnesia from magnesite, Ceram. Int., 48(23, Part A) (2022) 35525-35535.
[Google Scholar ↗](#)
- [4] L. Kong, Z. Deng, L. Cheng, M. Zhu
Reaction behaviors of Al-killed medium-manganese steel with glazed MgO refractory
Metall. Mater. Trans. B, 49 (6) (2018), pp. 3522-3533
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [5] B. Yang, Z. Deng, M. Zhu, Study on high-temperature reactions of MgO refractory with Ti-bearing Al-killed steel, Steel Re. Int., 95(7) (2024) 2400216.
[Google Scholar ↗](#)
- [6] Q. Wang, F. Tan, Z. He, G. Li, J. Li
Quantitative evaluation of slag corrosion on MgO-C refractory by experimental and numerical simulation
JOM, 73 (9) (2021), pp. 2709-2714
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [7] J. Cui, G. Yu, E. Wang, Y. Shen, T. Yang, K.C. Chou, X. Hou, Simulation and optimization of hydrogen-enriched shaft furnace, *cScience* 1(1) (2025) e1.
<https://doi.org/10.1002/csc3.1> ↗
[Google Scholar](#) ↗
- [8] X. Ren, B. Ma, J. Tang, Y. Li, J. Yu
Fabrication of energy-saving MgO with large grain size and low thermal conductivity: towards a new type of magnesia for high-temperature furnaces
Constr. Build. Mater., 342 (2022), Article 128097
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [9] S. Wang, L. Yu, Y. Guo, Y. Jiang, F. Chen, L. Yang
Viscosity and structural characterization of high-alumina CaO-SiO₂-MgO-Al₂O₃ slag for smelting of low-grade complex iron ore
JOM (2025), [10.1007/s11837-025-08034-2](https://doi.org/10.1007/s11837-025-08034-2) ↗
[Google Scholar](#) ↗
- [10] H. Kimura, T. Ogawa, M. Kakiki, A. Matsumoto, F. Tsukihashi
Effect of Al₂O₃ and MgO additions on liquidus for the CaO-SiO₂-FeO_x system at 1573 K
ISIJ Int., 45 (4) (2005), pp. 506-512
[Crossref](#) ↗ [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [11] L. Zhang, W.-L. Wang, H.-Q. Shao
Review of non-reactive CaO-Al₂O₃-based mold fluxes for casting high-aluminum steel
J. Iron Steel Res. Int., 26 (4) (2019), pp. 336-344
[Crossref](#) ↗ [Google Scholar](#) ↗
- [12] T. Wu, S. He, L. Zhu, Q. Wang

Study on reaction performances and applications of mold flux for high-aluminum steel

Mater. Trans., 57 (1) (2016), pp. 58-63

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [13] S.A. Nightingale, B.J. Monaghan, G.A. Brooks
Degradation of MgO refractory in CaO-SiO₂-MgO-FeO_x and CaO-SiO₂-Al₂O₃-MgO-FeO_x slags under forced convection

Metall. Mater. Trans. B, 36 (4) (2005), pp. 453-461

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [14] M.A.L. Braulio, M. Rigaud, A. Buhr, C. Parr, V.C. Pandolfelli
Spinel-containing alumina-based refractory castables

Ceram. Int., 37 (6) (2011), pp. 1705-1724



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [15] A. Huang, Y. Wang, Y. Zou, H. Gu, L. Fu
Dynamic interaction of refractory and molten steel: corrosion mechanism of alumina-magnesia castables

Ceram. Int., 44 (12) (2018), pp. 14617-14624



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [16] R. Sarkar, H.Y. Sohn
Interactions of alumina-based and magnesia-based refractories with iron melts and slags: a review

Metall. Mater. Trans. B, 49 (4) (2018), pp. 1860-1882

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [17] J. Chen, L. Huang, L. Dong, H. Zhang, Z. Huang, F. Li, S. Zhang
Hydration behavior of MgO surface: a first-principles study

Appl. Surf. Sci., 611 (2023), Article 155441



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [18] A. Valenzuela-Gutiérrez, J. López-Cuevas, A. González-Ángeles, N. Pilalua-Díaz
Addition of ceramics materials to improve the corrosion resistance of alumina refractories
SN Appl. Sci., 1 (7) (2019), p. 784
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [19] C. Reynaert
Corrosion tests for refractory materials intended for the steel industry - a review
Ceram. Silik. (2020), pp. 278-288
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [20] H. Yao, J. Liu, M. Xu, J. Ji, Q. Dai, Z. You
Discussion on molecular dynamics (MD) simulations of the asphalt materials
Adv. Colloid Interface Sci., 299 (2022), Article 102565
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [21] E. Adabifiroozjaei, J.N. Hart, P. Koshy, D.R.G. Mitchell, C.C. Sorrell
Mullite-glass and mullite-mullite interfaces: analysis by molecular dynamics (MD) simulation and high-resolution TEM
J. Am. Ceram. Soc., 101 (1) (2018), pp. 428-439
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [22] C. Guo, X. Tang, Y.E. Chen, C. Gao, Q. Shan, H. Wei, X. Liu, C. Lu, M. Fu, E. Wang, X. Liu, X. Hou, Y. Hou
From microstructure to performance optimization: innovative applications of computer vision in materials science
Int. J. Miner. Metall. Mater., 33 (1) (2026), pp. 94-115, [10.1007/s12613-025-3217-4 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] B. Kirchner, P.J. di Dio, J. Hutter

Real-world predictions from Ab initio molecular dynamics simulations

B. Kirchner, J. Vrabec (Eds.), Multiscale Molecular Methods in Applied Chemistry, Springer

Berlin Heidelberg, Berlin, Heidelberg (2012), pp. 109-153

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [24] R. Iftimie, P. Minari, M.E. Tuckerman
Ab initio molecular dynamics: concepts, recent developments, and future trends

Proc. Natl. Acad. Sci., 102 (19) (2005), pp. 6654-6659

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [25] P. Hohenberg, W. Kohn
Inhomogeneous electron gas

Phys. Rev., 136 (3B) (1964), pp. B864-B871

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [26] R. Car, M. Parrinello
Unified approach for molecular dynamics and density-functional theory

Phys. Rev. Lett., 55 (22) (1985), pp. 2471-2474

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [27] C.-M. Tang, X.-X. Chen, X.-L. Cheng, C.-Y. Zhang, Z.-P. Lu
Ab initio molecular dynamics study of thermite reaction at Al and CuO nano-interfaces at different temperatures

Chem. Phys. Lett., 699 (2018), pp. 171-181



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [28] Z. Li, Y. Feng, Y. Wen, X. Peng, Z. Cai, C. Peng, R. Wang
Ab initio molecular dynamics study on the local structures and solid/liquid interface in liquid Al-Ti and Al-B-Ti alloys

Mater. Today Commun., 39 (2024), Article 109290



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [29] C. Jiang, K. Li, J. Zhang, Q. Qin, Z. Liu, M. Sun, Z. Wang, W. Liang
Effect of MgO/Al₂O₃ ratio on the structure and properties of blast furnace slags: a molecular dynamics simulation
J. Non Cryst. Solids, 502 (2018), pp. 76-82
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [30] C. Fang, Z. Fan
Prenucleation at the liquid-Al/ α -Al₂O₃ and the liquid-Al/MgO interfaces
Comput. Mater. Sci, 171 (2020), Article 109258
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] S. Niu, Z. Yang, Y. Wang, D. Wang
Wetting and interfacial reaction of liquid CuAg-Ti alloy on SiO₂-BN ceramic
Appl. Surf. Sci., 576 (2022), Article 151878
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] X.H. Fan, B. Xu, Y. Xu, J. Li, L. Shi, F.M. Wang, J.P. Lin
Application of materials studio modeling in crystal structure
Adv. Mat. Res., 706–708 (2013), pp. 7-10
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [33] Y.-X. Wang, Y.-G. Li, Y.-B. Gao, Z. Huang, H.-J. Zhang
High-temperature wetting behavior between slag and refractory
J. Iron Steel Res. Int., 31 (6) (2024), pp. 1304-1319
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [34] L. Cheng, L. Zhang, Y. Ren
Wettability between 304 stainless steel and refractory materials
J. Mater. Res. Technol., 9 (3) (2020), pp. 5784-5793
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [35] G. Cheng, T. Zhao, Y. Zhao, T. Jia, Y. Li, X. Wang, C. Dong
Wetting, cladding and corrosion properties of alkaline slag on dense corundum refractories
Ceram. Int., 48 (4) (2022), pp. 5795-5804
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [36] J. Morgiel, N. Sobczak, M. Pomorska, R. Nowak
First stage of reaction of molten Al with MgO substrate
Mater Charact, 103 (2015), pp. 133-139
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [37] V.G. Tsirelson, A.S. Avilov, Y.A. Abramov, E.L. Belokoneva, R. Kitaneh, D. Feil
X-ray and electron diffraction study of MgO
Acta Crystallogr. B, 54 (1) (1998), pp. 8-17
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [38] L.D. Madsen, R. Charavel, J. Birch, E.B. Svedberg
Assessment of MgO(100) and (111) substrate quality by X-ray diffraction
J. Cryst. Growth, 209 (1) (2000), pp. 91-101
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [39] X. Liu, L. Dong, Y. Xing, L. Huang, H. Zhang, S. Zhang
Interaction between MgO and molten Fe: roles of crystal orientation, surface defects, and oxygen in melt
Surf. Interfaces, 73 (2025), Article 107617
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [40] X. Liu, L. Dong, L. Huang, H. Zhang, S. Zhang
Atomic-scale insights into high-temperature interfacial interactions between molten Al and MgO
J. Am. Ceram. Soc., 109 (1) (2026), Article e70305
[View in Scopus ↗](#) [Google Scholar ↗](#)

- [41] C. Fang, Z. Fan
Atomic ordering at the interfaces between liquid Al and solid MgO: an Ab initio molecular dynamics study
Philos. Mag. Lett., 100 (5) (2020), pp. 235-244
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [42] M.Z. Hossain
Interface induced band gap reduction and electron transfer in semiconducting carbon nanotube
Appl. Phys. Lett., 96 (5) (2010), Article 053118
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [43] Y. Hinuma, G. Pizzi, Y. Kumagai, F. Oba, I. Tanaka
Band structure diagram paths based on crystallography
Comput. Mater. Sci, 128 (2017), pp. 140-184
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [44] W.S. Morgan, J.E. Christensen, P.K. Hamilton, J.J. Jorgensen, B.J. Campbell, G.L.W. Hart, R.W. Forcade
Generalized regular k-point grid generation on the fly
Comput. Mater. Sci, 173 (2020), Article 109340
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [45] M.A.L. Marques, M.J.T. Oliveira, T. Burnus
Libxc: a library of exchange and correlation functionals for density functional theory
Comput. Phys. Commun., 183 (10) (2012), pp. 2272-2281
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [46] G. Kresse, J. Hafner
Norm-conserving and ultrasoft pseudopotentials for first-row and transition elements

J. Phys. Condens. Matter, 6 (40) (1994), p. 8245

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [47] P.E. Blöchl
Projector augmented-wave method

Phys. Rev. B, 50 (24) (1994), pp. 17953-17979

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [48] S. Nosé
A unified formulation of the constant temperature molecular dynamics methods

J. Chem. Phys., 81 (1) (1984), pp. 511-519

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [49] W.G. Hoover
Canonical dynamics: equilibrium phase-space distributions

Phys. Rev. A, 31 (3) (1985), pp. 1695-1697

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [50] H.J. Monkhorst, J.D. Pack
Special points for brillouin-zone integrations

Phys. Rev. B, 13 (12) (1976), pp. 5188-5192

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [51] A. Koryttseva, A. Navrotsky
High-temperature calorimetric study of oxide component dissolution in a CaO–MgO–Al₂O₃–SiO₂ slag at 1450°C

J. Am. Ceram. Soc., 100 (3) (2017), pp. 1172-1177

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [52] J. Hautman, M.L. Klein
Microscopic wetting phenomena

Phys. Rev. Lett., 67 (13) (1991), pp. 1763-1766

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [53] Y.Z. Povstenko
Anisotropy of wetting and spreading

J. Sov. Math., 64 (3) (1993), pp. 890-898

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [54] G. Henkelman, A. Arnaldsson, H. Jónsson
A fast and robust algorithm for Bader decomposition of charge density

Comput. Mater. Sci, 36 (3) (2006), pp. 354-360



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [55] R.F.W. Bader
Atoms in molecules

Acc. Chem. Res., 18 (1) (1985), pp. 9-15

[Crossref ↗](#) [Google Scholar ↗](#)

- [56] J.B. Goodenough
First-order changes in ionic/covalent bonding

Ferroelectrics, 130 (1) (1992), pp. 77-86

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [57] E.G. Shafrin, W.A. Zisman
Constitutive relations in the wetting of low energy surfaces and the theory of the retraction method of preparing monolayers1

J. Phys. Chem., 64 (5) (1960), pp. 519-524

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [58] J.Y.Y. Heng, A. Bismarck, A.F. Lee, K. Wilson, D.R. Williams
Anisotropic surface energetics and wettability of macroscopic form I paracetamol crystals

Langmuir, 22 (6) (2006), pp. 2760-2769

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [59] J.-H. Park, I.-H. Jung, H.-G. Lee
Dissolution behavior of Al₂O₃ and MgO inclusions in the CaO-Al₂O₃-SiO₂ slags: formation of ring-like structure of MgAl₂O₄ and Ca₂SiO₄ around MgO inclusions

ISIJ Int., 46 (11) (2006), pp. 1626-1634

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

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